Question	Answer	Marks
1	**Defining the problem**.	
	f is the independent variable and *E* is the dependent variable **or** vary *f* and measure *E*	**1**
	keep *V* and *R* constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • coils C and D placed with their axes on a straight line • separate workable circuit for coil D • (a.c.) voltmeter or oscilloscope connected across coil D (Do not accept a power supply connected to coil D.)	**1**
	a.c. power supply/signal generator connected to coil C	**1**
	workable circuit for coil C with power supply and (a.c.) voltmeter/oscilloscope in parallel with resistor and coil C	**1**
	method to determine *f*, e.g. read from signal generator or use of oscilloscope	**1**
	Method of Analysis	
	plot a graph of lg *E* against lg *f* or equivalent (e.g. ln *E* against ln *f*)	**1**
	q = gradient	**1**
	$p = \frac{R}{V} \times 10^{y\text{-intercept}}$ (for ln *E* against ln *f*: $p = \frac{R}{V} \times e^{y\text{-intercept}}$)	**1**

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Question	Answer	Marks
1	**Additional detail including safety considerations**	**6**
	D1 precaution (to prevent burns) from hot coils/hot resistor, e.g. use gloves to handle hot coil/resistor, switch off circuit and wait for hot coil/resistor to cool	
	D2 keep the number of turns on each coil constant	
	D3 keep distance between the coils constant	
	D4 workable circuit diagram to determine R. e.g. circuit with ammeter connected in series and voltmeter in parallel with resistor **or** resistor connected to ohmmeter only	
	D5 determination of resistance R: potential difference across $R \div$ current in R **or** use ohmmeter to measure R	
	D6 method to keep distance between the coils constant, e.g. fix/clamp coils to bench	
	D7 method to determine f from oscilloscope, e.g. period T = time-base $\times$ horizontal distance **and** $f = 1 / T$	
	D8 method to determine V or E from oscilloscope, e.g. V or E = y-gain $\times$ vertical distance	
	D9 method to increase E e.g. use iron core, place coils closer, increase V, decrease R	
	D10 relationship valid if a straight line is produced (passing through log $\left(\frac{pV}{R}\right)$) Do not accept line passing through the origin.	

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